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RESEARCH INTERESTS

Individualized brain topography · Precision functional mapping · Gene–environment interplay · Brain dynamics
Computational modeling · Internalizing psychopathology · Risk and resilience · Adolescent development

EDUCATION

Virginia Tech, Blacksburg, USA — Aug 2025 – Present
Ph.D. Student, Developmental Science, Department of Psychology
Advisor: Prof. Tae-Ho Lee

Hangzhou Normal University, Hangzhou, China — Sep 2021 – Jul 2024
M.A. in Applied Psychology
Advisor: Prof. Bao-ming Li

The University of British Columbia, Vancouver, Canada — Jun 2023 – Aug 2023
Visiting Master's Student, Overseas Study Program, Faculty of Education

Henan University, Kaifeng, China — Sep 2017 – Jul 2021
B.S. in Applied Psychology

RESEARCH GRANTS

Seed Grant Program for Scientific Research, Hangzhou Normal University — 2022–2023
Role: Principal Investigator
Title: A Questionnaire and Brain Imaging Study on the Relationship Between Stressful Life Events and Sleep in University Students

Pilot Study, STI2030-Major Projects (China Brain Project) — 2022–2023
Role: Pilot Study Lead
Title: Cognitive, Emotional, and Intellectual Profiles of Children with Long-term Weiqi (Go) Training

General Research Project, Zhejiang Provincial Department of Education — 2022–2024
Role: Co-Investigator
Title: Effects of Computerized Working Memory Training on Individual Proactive and Reactive Control

PUBLICATIONS

Peer-Reviewed Journal Articles

[1] **Li, Q.**, Ye, X., Li, Z., Yang, S., Yin, L., Li, B.-M., & Wang, C. (2024). Association between stressful life events and sleep quality in Chinese university students: Mediating and moderating roles of emotion regulation. *Journal of Health Psychology*, *30*(5), 909–920. [DOI]

[2] Ren, T., Ye, X., Li, Z., **Li, Q.**, Zhang, X., Dou, W., Jia, X., Li, B.-M., & Wang, C. (2023). Associations between physical activity and proactive control and the modulating role of working memory. *Psychology of Sport and Exercise*, *66*, 102374. [DOI]

Manuscripts Under Review

[3] **Li, Q.*,** Ye, Y.*, Ren, T., Xue, M., Lee, T.-H., & Li, B.-M. (2026). Cumulative stress contextualizes associations between vmPFC–limbic connectivity and sleep quality across psychological resilience profiles [Major revision at *Biological Psychology*, special issue "Resilience in the Biological–Psychological–Social Cascade"]. *Equal contribution.

[4] **Li, Q.**, Chen, Y.-Y., Kim-Spoon, J., Casas, B., Qu, Y., & Lee, T.-H. (2026). Parent–child cortical similarity indexes multiple dimensions of sensitivity in the association between family environment and adolescent internalizing symptoms [Under review at American Psychologist]. *bioRxiv*. [DOI]

[5] **Li, Q.**, Chen, Y.-Y., Zhou, Z., Qu, Y., Kim-Spoon, J., & Lee, T.-H. (2026). A convergent behavioral–neural profile of parental dysregulation and the link between parent–child brain-state similarity and child negative affect [Under review at Human Brain Mapping]. *bioRxiv*. [DOI]

[6] Lee, T.-H., Chen, Y.-Y., **Li, Q.**, Yang, B., Zhou, Z., & Qu, Y. (2026). Family conflict and parent–child similarity in affective valuation [Under review]. *bioRxiv*. [DOI]

[7] Lv, X., Wu, Z., Zhang, Z., Xue, M., Guo, J., Liu, S., **Li, Q.**, Liu, B., He, X., Xi, Y., Wang, K., Chen, Z., & Chen, R. (2026). A multimodal neuroimaging large language model for transdiagnostic suicidal risk assessment across developmental stages [Under review at *Nature Machine Intelligence*].

Manuscripts in Preparation

[8] **Li, Q.**, et al. (in preparation). Baseline levels of daily affect, not affective dynamics, predict suicidal ideation in college students two months later: Evidence from hidden Markov models and external validation

[9] **Li, Q.**, et al. (in preparation). Partitioning a child's individualized network topography into species-shared, parent-shared, and child-specific parts: Evidence from precision functional mapping of parent–child dyads.

[10] Chen, Y.-Y., Kim, H., **Li, Q.**, & Lee, T.-H. (in preparation). Shared and distinct neural responses during aggressive and prosocial behaviors: A systematic review of fMRI studies across adolescence and young adulthood.

CONFERENCE PRESENTATIONS

[11] **Li, Q.**, Chen, Y.-Y., Zhou, Z., Qu, Y., Kim-Spoon, J., & Lee, T.-H. (2027, April). *Parent–child neural similarity in the link between family environment and adolescent adjustment* [Symposium paper proposal submitted]. Society for Research in Child Development Biennial Meeting, Atlanta, GA, USA.

[12] **Li, Q.**, Chen, Y.-Y., & Lee, T.-H. (2026, November). *Parent internalizing shapes the link between parent–child latent brain-state similarity and child recent negative affect* [Nanosymposium talk]. Society for Neuroscience Annual Meeting, Washington, DC, USA. (Trainee Professional Development Award)

[13] **Li, Q.**, Chen, Y.-Y., & Lee, T.-H. (2026, March). *A topological principle of parent–child brain morphological similarity* [Poster and Data Blitz presentation]. Cognitive Neuroscience Society Annual Meeting, Vancouver, BC, Canada. (Graduate Student Award)

[14] **Li, Q.**, Zhang, Y., Li, B.-M., & Li, Z. (2023, October). *The effect of expectation on the regulatory mechanisms of cognitive control* [Oral presentation]. 25th National Academic Congress of Psychology, Chinese Psychological Society, Chengdu, China.

[15] Yang, J., Zhang, Y., **Li, Q.**, Li, B.-M., & Li, Z. (2023, October). *The influence of information fuzziness and group pressure on conformity behavior and its neural mechanism* [Oral presentation]. 25th National Academic Congress of Psychology, Chinese Psychological Society, Chengdu, China.

[16] **Li, Q.**, Ye, X., Li, B.-M., & Wang, C. (2023, October). *The mediating role of emotional regulation in the associations of stressful life events with sleep quality and depressive experience in Chinese university students* [Poster presentation]. 25th National Academic Congress of Psychology, Chinese Psychological Society, Chengdu, China.

RESEARCH EXPERIENCE

Note. Numbers in brackets refer to entries in Publications and Conference Presentations.

Graduate Researcher, Affective Neuroscience and Development Lab, Virginia Tech **Aug 2025 – Present**

Advisor: Prof. Tae-Ho Lee

- **Lead project · Cortical morphometric similarity**: Asked whether sensitivity to family environment resides in the parent–child relationship rather than in the child alone; separated regional from organizational cortical similarity (400-parcel morphometry, morphometric similarity networks; 115 dyads, two cohorts); conceived the question and led the analysis and writing [4, 11, 13].

- **Lead project · Brain-state dynamics (HMM)**: Tested whether parent–child brain-state similarity is protective or risky depending on the parent; modeled time-resolved brain states during an emotional film (HMM-MAR, neural index of parental state dynamics; 32 dyads); led the analysis and writing [5, 12].
- **Lead project · Precision functional mapping**: Set out to resolve where, how much, and in what unit genes and environment are jointly passed from parent to child in the brain; partitioned each child's network topography into species-shared, parent-shared, and child-specific parts (MS-HBM, variance partitioning, dyadic identifiability; 28 dyads); conceived the question and led the design, analysis, and writing [9].
- **Pipeline lead · 8-year longitudinal adolescent morphometry**: Asked how cortical structure changes across adolescence; built a FreeSurfer longitudinal pipeline extracting thickness, surface area, curvature, and gyrification across waves (with Prof. Jungmeen Kim-Spoon); led pipeline development [4].
- **Self-initiated project · Precision self-study**: Asked how my own brain connects to my everyday emotions, behavior, and decisions; mapped my individualized networks from 2 h 40 min of my own fMRI and linked them to one month of EMA, a year-long diary, questionnaires, and computational models of my task behavior; designed and ran every part.
- **Side projects · Systematic review and manuscript editing**: Co-authored a systematic review on aggressive and prosocial behavior; edited a manuscript on family conflict and parent–child affective valuation [6, 10].

Collaborative Project (Remote), University of Notre Dame **Jun 2026 – Present**

Precision Brain Mapping Lab; Supervised by Dr. Jingnan Du

- **Pipeline lead · Precision functional mapping at 7T**: Deployed the iProc pipeline (Buckner Lab, Harvard) on Notre Dame's computing cluster and applied it to densely sampled 7T data (Natural Scenes Dataset); led the deployment.
- **Lead analyst · Individualized language networks**: Asked whether two nested language networks replicate in an independent 7T dataset; estimated individual networks with MS-HBM and verified boundaries with seed-based connectivity; led the analysis.
- **Lead analyst · Cortico-subcortical mapping**: Asked whether the cortical dissociation holds in cerebellum, thalamus, and striatum; mapped subcortical voxels to individualized cortical networks; led the analysis.
- **Method transfer · Parent–child dyadic precision functional mapping**: Adapted the individual-level mapping workflow for parent–child dyads [9].

Collaborative Project (Remote): E-MIND, Multi-institutional **Sep 2026 – Present**

Ecological Mechanisms in Young Adults' Mental Health
Supervised by Prof. Mark Shuquan Chen (Harvard University), Prof. Shufang Sun (Brown University), and Prof. Bin Yu (Wuhan University)

- **Lead analyst · Affect-dynamics modeling (EMA, HMM)**: Examined whether how affect moves adds information beyond where it sits; modeled 34 affect-dynamics indices, including hierarchical HMMs, in a two-university EMA study (N = 2,549); led all analyses.
- **Lead analyst · Prediction and validation (ML)**: Asked whether apparent dynamics reflect genuine signal or artifacts of mean levels; built cross-validated, externally validated prediction models and null simulations; led all analyses.
- **Lead project · Daily affect and suicidal ideation**: Leading a first-author manuscript on which features of daily affect predict later suicidal ideation [8].

Research Assistant, Youth Mental Health Lab, Tsinghua University **May 2024 – Jun 2025**

Advisors: Prof. Runsen Chen, Dr. Zhijun Wu, & Dr. Diyang Qu

- **Independent analysis · Multimodal MRI in adolescent self-injury**: Asked how brain structure and connectivity differ in adolescents with NSSI and emotion-regulation difficulties; analyzed gray matter and static and dynamic connectivity (Peking University Sixth Hospital cohort); led the analysis independently.

- **Analyst · Preliminary analyses in large cohorts (A-HELP, ABCD)**: Ran preliminary linear mixed models in the ABCD Study ($N \approx 11{,}800$) linking pre-computed polygenic risk scores and subregional gray matter volume to NSSI, and latent profile and network analyses in the A-HELP cohort ($N = 94{,}412$; 393 schools) of family environment, emotion regulation, and personality; findings reported to the PI.
- **Co-designer · Neuromodulation (taVNS) study**: Co-designed a cross-site study of transcutaneous auricular vagus nerve stimulation (taVNS) mechanisms in adolescents; drafted the IRB protocol.
- **Pipeline builder · MRI preprocessing**: Built the lab's BIDS-based preprocessing pipeline and ran resting-state connectivity and VBM analyses for 2 MRI datasets.
- **Data collection · Multimodal clinical assessment**: Collected interview, EMA, EEG, and fMRI data and trained six interns [7].

Master's Student Researcher, Hangzhou Normal University **Sep 2021 – May 2024**

Institute of Brain Science; Advisors: Prof. Bao-ming Li, Prof. Chunjie Wang, Dr. Zhenghan Li, & Dr. Zhong Yang

- **Grant PI · Stress, sleep, and resting-state fMRI**: Led a questionnaire and MRI study of stressful life events and sleep as PI of a university grant: collected longitudinal survey data from 1,169 university students and multimodal MRI (T1, resting-state, DTI) from 64 participants; tested mediating and moderating roles of emotion regulation and ROI-based amygdala and medial prefrontal connectivity [1, 3, 16].
- **Pilot lead · Long-term Weiqi training in children**: Led a pilot study on cognitive, emotional, and intellectual profiles of children with long-term Weiqi (Go) training; programmed E-Prime tasks (Raven's matrices, Go/No-Go, cued task switching) and assessed 159 children and adolescents of Raven's Progressive Matrices, cognitive tasks and socioemotional questionnaires.
- **Study lead · EEG and eye-tracking**: Designed and ran an EEG and eye-tracking study (Neuroscan, Tobii, E-Prime; $n = 64$) on how expectation and statistical learning regulate attention to targets and distractors [14, 15].
- **Co-Investigator · Working memory training**: Co-investigator on a provincial grant studying proactive and reactive control: reviewed the literature for the grant application, adapted AX-CPT, N-back, and backward digit span tasks, and preprocessed structural and resting-state MRI (SPM, DPABI, CONN) [2].

TEACHING EXPERIENCE

Virginia Tech, Department of Psychology Blacksburg, VA
Graduate Teaching Assistant, Introduction to Psychology Spring 2026 – Present
Graduate Teaching Assistant, Developmental Psychology Fall 2025

Tianyuan College Hangzhou, China
Intern Mental Health Teacher (Grades 4–6) Sep 2023 – Dec 2023

167 PSYCLASS Hangzhou, China
Founder and Tutor, Master's Entrance Exam Preparation in Psychology Sep 2021 – Feb 2022

Jiying Primary School Kaifeng, China
Trainee Mental Health Teacher (Grade 1) Sep 2020 – Dec 2020

MENTORING AND SERVICE

Graduate Application Mentor (Pro Bono)
PsyCN2none (Team Lead) 2025–Present
1HourProject Fall 2025
NPSP Fall 2025

- Provide one-on-one mentoring to students applying to graduate programs in psychology, including program selection and review of statements of purpose, personal statements, and CVs.
- Coordinate a volunteer team of 17 mentors supporting 10 mentees: organize application workshops, set the timeline, and track progress and feedback across the application cycle.

"Right to Play" (NGO), Kaifeng Children's Hospital Kaifeng, China
Part-time Play Coach Oct 2018 – Dec 2019

- Used therapeutic play and picture-book preparation to reduce distress in chronically ill children and children awaiting surgery.

SKILLS

Statistical Methods: Multiple regression, mediation and moderation analysis, structural equation modeling, cross-lagged panel models, growth modeling, latent class/profile analysis, network analysis, linear mixed and multilevel models, hidden Markov models (including hierarchical HMM)

Machine Learning: Penalized regression, gradient boosting, cross-validation, external validation, calibration, decision-curve analysis, connectome-based predictive modeling

Neuroimaging Methods: Task activation; static and dynamic functional connectivity; seed-based connectivity; graph-theoretic network analysis; effective connectivity; HMM-MAR brain-state analysis; voxel- and surface-based morphometry; morphometric similarity; precision functional mapping (individualized parcellation, network variants, variance partitioning); cortico-subcortical mapping

Neuroimaging Software: FreeSurfer, fMRIPrep, Connectome Workbench, SPM, CAT12, DPABI, CONN

Programming and Statistics: R, MATLAB, Mplus, SPSS; Linux high-performance computing clusters

Data Collection: E-Prime; eye tracking (Tobii); EEG (Neuroscan, Brain Products, Curry 8); MRI (GE Discovery MR750 3T, Siemens Prisma 3T); EMA; semi-structured clinical interviews

Languages: Chinese (native), English (proficient)

Certification: Teacher Qualification Certificate, Senior High School Psychology (China)

SCHOLARSHIPS AND AWARDS

Trainee Professional Development Award, Society for Neuroscience (USD 1,000) 2026
Graduate Student Award, Cognitive Neuroscience Society (USD 100) 2026
Academic Scholarship, Hangzhou Normal University (CNY 9,000) 2021–2023
School Outstanding Student Cadre, Hangzhou Normal University (CNY 1,000) 2023
Prize for Community Work, Hangzhou Normal University (CNY 1,000) 2022–2023
Excellent Graduate Student, Hangzhou Normal University (CNY 1,000) 2022–2023
Practical Service Award, Hangzhou Normal University (CNY 1,000) 2022–2023
Exchange Student Fellowship, Hangzhou Normal University (CNY 120,000) 2023
Second Prize, National Applied Psychological Practice Skills Competition, Hangzhou Normal University 2022
Third Prize, "Challenge Cup" National College Student Curricular Academic Science and Technology Works Competition, Henan University 2020

MEDIA COVERAGE

Research highlight of preprint [4], Virginia Tech Institute for Society, Culture and Environment 2026
Feature on preprint [4], PsyBrain Advance (WeChat official account; in Chinese) 2026
News feature on article [1], Institute of Brain Science, Hangzhou Normal University (in Chinese) 2024

MEMBERSHIPS

Society for Neuroscience 2026–Present
Society for Research in Child Development 2026–Present
Cognitive Neuroscience Society 2025–Present
Association of Chinese Helping Professionals and Psychologists–International 2024–Present
Chinese Psychological Society 2022–Present
Organization for Human Brain Mapping 2021–2022